

SB-GRPO: Semantic-Balanced GRPO for Enhanced LLM Reasoning

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Abstract. Reinforcement Learning (RL), specifically Group Relative Policy Optimization (GRPO), has emerged as a transformative technique for unlocking deep reasoning in Large Language Models (LLMs). By incentivizing multi-step Chain-of-Thought (CoT) behaviors, GRPO allows models to transcend simple pattern matching. However, current implementations are plagued by critical bottlenecks: advantage vanishing in uniform groups, systematic correction bias during reflection, and inefficient utilization of noisy, high-entropy error samples. We propose SB-GRPO (Semantic-Balanced GRPO), an advanced framework that integrates dynamic multi-layer reflection with semantic error profiling. Our architecture introduces a differentiable Heuristic Confidence Gate that smoothly transitions between standard and negative-enhanced advantage calculation, preventing gradient stalls. We further employ K-Means clustering on reasoning embeddings to identify systematic logical failure modes and construct Static Balanced Batches to stabilize policy gradients. Finally, we derive an information-theoretic loss scaling factor based on the Shannon Entropy of semantic clusters to prioritize fixing systematic errors over random noise. Evaluated under strict parameter-efficient fine-tuning constraints (4-bit QLoRA) using a 7B model, SB-GRPO achieves a highly competitive pass@1 accuracy of 74.80% on the MATH-500 dataset, yielding an absolute improvement of 26.40% over vanilla GRPO. Compared to M-GRPO, a state-of-the-art multi-layer baseline, our method delivers a 2.20% overall gain, with pronounced improvements in complex domains such as Precalculus (+5.68%) and Intermediate Algebra (+4.88%). Furthermore, it exhibits substantial gains in elite logical reasoning, achieving 10.00% (3/30) accuracy on the challenging AIME 2024 benchmark in single-pass evaluation, and an average of 12.50% across multiple evaluation runs, significantly outperforming the 3.33

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1 Introduction

The paradigm of Large Language Model (LLM) training has undergone a pivotal shift from Supervised Fine-Tuning (SFT)[16] to Reinforcement Learning-based (RL)[8] alignment. While models such as GPT-4 [1] and Claude [2] have demonstrated outstanding reasoning capabilities, their reliance on complex reward models imposes significant computational and scalability bottlenecks. Recently, the release of DeepSeek-R1 popularized Group Relative Policy Optimization (GRPO)[10] [20], marking a milestone in training specialized reasoning models without the heavy overhead of a traditional reward model. By leveraging group-relative rewards, GRPO allows the policy model to autonomously distinguish between good and bad reasoning paths within the same prompt context, effectively optimizing for complex system thinking capabilities.

However, as models scale and tasks become more complex, such as Olympiad-level mathematics and complex reasoning chains, vanilla GRPO faces diminishing returns. We identify three fundamental limitations in existing RL pipelines for reasoning:

1. Zero Advantage: In many reasoning tasks, especially at the start or late stages of training, a model may generate rollouts that all share the same outcome, either all correct or all incorrect. In such cases, the mean reward equals the individual reward, and the standard deviation approaches zero, causing the advantage signal to become zero [22]. This stalls the gradient and wastes expensive compute.

2. The Compulsive Correction Fallacy: Multi-layer [7] or iterative reflection models [14] often suffer from overcorrection. When prompted to reflect, models exhibit a systematic bias towards altering their initial answers, even when correct. This bias degrades the stability of the model outputs and leads to a high rate of false-negative revisions.

3. The Noise-Signal Dichotomy: Traditional RL update paradigms implicitly operate under the assumption that all optimization errors possess equivalent informational value. However, a stochastic error originating from a superficial computational oversight yields significantly less pedagogical utility than a structural error rooted in a fundamental logical misconception. Standard GRPO frameworks currently lack the granular mechanism required to disambiguate these distinct failure modes.

To address these limitations, we propose Semantic-Balanced GRPO (SB-GRPO). Our framework tackles the zero-advantage problem by introducing a Heuristic Confidence Gate, which applies a differentiable switching mechanism to activate the Advantage Calibration mechanism introduced in [15], which we formulate as a Calibrated Advantage (A_{calib}), using virtual maximum rewards for uniform batches. To mitigate correction bias and differentiate noise from systematic errors, we integrate a Semantic Error Profiling (SEP) module within a two-layer generation-reflection pipeline. By clustering reasoning paths and constructing Static Balanced Batches, our approach ensures that the model optimizes based on the most highly informative errors rather than trivial computational noise.

2 Related Work

Reinforcement Learning from Human Feedback (RLHF) [4] traditionally relies on Proximal Policy Optimization (PPO) [19], which suffers from heavy computational overhead due to separate reward models. Group Relative Policy Optimization (GRPO) [20] circumvents this bottleneck by substituting the critic with a baseline derived from the average rewards of G distinct rollouts. This memory-efficient paradigm was further refined by the Advantage Calibration mechanism introduced in [15], which incorporates virtual maximum rewards to maintain continuous gradient flow in extreme edge cases. Concurrently, augmenting LLM reasoning via Chain-of-Thought [21] and autonomous self-correction [14] has gained significant traction. Building on this, Multi-Layer GRPO (MGRPO) [7] employs a two-layer pipeline that utilizes initial outputs as context to verify and correct errors. However, such unconstrained self-reflective frameworks often induce a pronounced correction bias, coercing models to arbitrarily alter valid paths and thereby degrading inference stability.

To mitigate reasoning instability and overcorrection, recent studies focus on uncertainty quantification and sophisticated data curation. Semantic Entropy (SE) [13] provides an information-theoretic metric to distinguish genuine model confusion from generative diversity—a concept SEED-GRPO [3] leverages to dynamically dampen gradient updates for highly uncertain prompts. Additionally, theoretical insights from Balanced Sampling [9] emphasize that maintaining an equitable distribution of correct and hard-negative samples is pivotal for stabilizing convergence in sparse-reward environments. Collectively, these advancements underscore the necessity of a unified framework. Building upon this foundation, our proposed approach combines semantic error profiling with statically balanced batches to robustly optimize iterative reasoning while mitigating correction bias.

3 Methodology

This section details the SB-GRPO architecture (Fig. 1). To resolve GRPO instability, we introduce a Heuristic Confidence Gate (Sec. 3.2) that dynamically routes uniform batches to use Advantage Calibration, preventing zero advantage. For mixed batches, Layered Generation and Reflection (Sec. 3.3) and Semantic Error Profiling (Sec. 3.4) cluster reasoning trajectories to target systematic failures. The gradients are modulated via an Information-Theoretic Loss Scaling factor (Sec. 3.5) based on error entropy. The step-by-step integration of these modules into a cohesive training iteration is formalized in Algorithm 1.

3.1 Formal Derivation of SB-GRPO

Let q be a prompt and $O = \{o_1, \dots, o_N\}$ be N rollouts from policy π_θ . The standard reward for o_i is $r_i \in \{0, 1\}$. In SB-GRPO, we define the accuracy ratio:

$$R = \frac{1}{N} \sum_{i=1}^N r_i$$

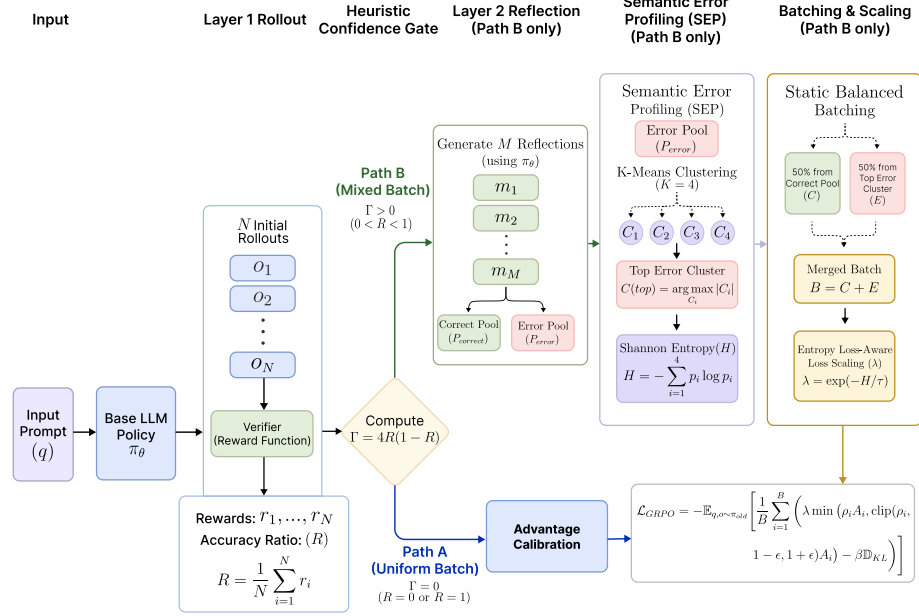


Fig. 1. The complete system architecture of SB-GRPO. The Heuristic Confidence Gate (Γ) dynamically routes uniform batches to the Advantage Calibration module (Path A) to prevent advantage vanishing. For mixed batches (Path B), the framework applies Multi-Layer Reflection and Semantic Error Profiling (SEP) to extract the Top Error Cluster (C_{top}), forming a Static Balanced Batch optimized by an Entropy-Aware Loss Scaling factor (λ).

3.2 Heuristic Confidence Gate

To maintain gradient continuity, we define a differentiable heuristic gating function $\Gamma(R)$ using the quadratic form:

$$\Gamma(R) = 4R(1 - R) \quad (1)$$

Note that $\Gamma \in [0, 1]$, reaching 0 at $R = 0$ (all wrong) or $R = 1$ (all correct), and 1 at $R = 0.5$. This parabolic form ensures that purely correct or incorrect batches seamlessly switch to Calibrated Advantage loss, while balanced batches optimally utilize multi-layer reflection without gradient spikes. This gate selects the advantage calculation method:

$$A_i = \Gamma \cdot A_{\text{std},i} + (1 - \Gamma) \cdot A_{\text{calib},i} \quad (2)$$

where A_{calib} uses a virtual reward $r_{\text{virtual}} = 1$ to inject a negative signal even when all $r_i = 0$. Injecting this virtual reward increases the mean μ , which in turn pushes the $A_{\text{std},i} = (r_i - \mu)/\sigma$ of incorrect samples into explicitly negative

values rather than leaving them stuck at 0. The standard advantage is computed as $A_{\text{std},i} = (r_i - \mu)/\sigma$, where μ and σ are the mean and standard deviation of rewards within the group. When variance approaches zero, A_{std} approaches zero, making A_{calib} critical for continuous training. Conversely, when $R = 1$ (all correct), injecting $r_{\text{virtual}} = 1$ maintains a zero advantage, correctly signaling that no policy update is required for an already mastered prompt.

To illustrate the practical effect of this adaptive mechanism, consider a sample size of $N = 8$:

- **Case 1: Homogeneous-Incorrect (0 correct, 8 wrong, $R = 0$).** The gate yields $\Gamma = 0$. Standard GRPO would provide $A_{\text{std}} = 0$ for all samples, halting the learning process. By routing purely to Advantage Calibration, our framework injects $r_{\text{virtual}} = 1$, producing an augmented mean $\mu' \approx 0.11$ and $\sigma' \approx 0.31$. This assigns a meaningful negative advantage $A \approx -0.35$ to all incorrect samples, forcing the model to explore new reasoning paths.
- **Case 2: Low-Accuracy Mixed (1 correct, 7 wrong, $R = 0.125$).** The gate yields $\Gamma = 0.4375$. Standard GRPO assigns an overly aggressive reward to the single correct sample ($A_{\text{std}} \approx +2.65$) and a weak penalty to the wrong ones ($A_{\text{std}} \approx -0.38$). Our dynamically routed advantage rebalances these to $A_{\text{correct}} \approx +2.21$ and $A_{\text{wrong}} \approx -0.47$. This dampens the variance of lucky guesses while increasing the penalty for failures, strongly promoting exploration when accuracy is low.
- **Case 3: High-Accuracy Mixed (7 correct, 1 wrong, $R = 0.875$).** The gate yields $\Gamma = 0.4375$. Standard GRPO computes $A_{\text{std}} \approx +0.38$ for correct and -2.65 for incorrect samples. Our routed advantage minimally shifts these to $+0.36$ and -2.75 , respectively. In this high-accuracy scenario, the mechanism makes minimal adjustments, stably favoring exploitation while maintaining a strict penalty for the rare failure.

3.3 Layered Generation and Reflection

Rather than relying on a single-pass generation, SB-GRPO isolates the exploration and verification phases into a two-layer pipeline. This deliberate separation allows the model to systematically audit its own reasoning paths before gradient updates are computed. The process is defined as follows:

Layer 1 (Rollout): $N = 8$ samples are generated with temperature $\tau = 0.7$ to explore the solution space.

Layer 2 (Reflection): If $\Gamma > 0$, the model performs $M = 2$ reflections per rollout using a neutral system prompt: *"Review the solution logic. Only correct it if errors are found; otherwise, maintain consistency."* This creates a pool of $N \times M = 16$ reasoning paths.

Algorithm 1 SB-GRPO Training Step

```

1: Input: Prompt  $q$ , Policy  $\pi_\theta$ , Verifier  $V$ 
2: Generate  $N$  rollouts  $O = \{o_1, \dots, o_N\}$  via vLLM
3:  $R \leftarrow \text{Accuracy}(O, V)$ 
4:  $\Gamma \leftarrow 4R(1 - R)$ 
5: if  $\Gamma > 0$  then
6:   Generate  $M$  reflections per rollout  $\rightarrow$  Pool  $P$ 
7:   Cluster  $P_{\text{error}}$  into  $\{C_1 \dots C_K\}$  using K-Means
8:    $\lambda \leftarrow \exp(-\text{Entropy}(C)/\tau)$ 
9:    $B \leftarrow \text{Sample } S/2 \text{ from } P_{\text{correct}} \text{ and } S/2 \text{ from } C_{\text{top}} \text{ (with replacement)}$ 
10: else
11:    $\lambda \leftarrow 1.0$ ,  $B \leftarrow \text{Sample } S \text{ uniformly from } O \text{ (with replacement)}$ 
12:   Use Calibrated Advantage  $A_{\text{calib}}$ 
13: end if
14: Training Phase:
15: Flush vLLM cache  $\rightarrow$  Init PyTorch Trainer
16:  $\rho \leftarrow \frac{\pi_\theta}{\pi_{\text{old}}}$ 
17:  $\mathcal{L}_{\text{surr}} \leftarrow \min(\rho A, \text{clip}(\rho, 1 - \epsilon, 1 + \epsilon) A)$ 
18:  $\mathcal{L}_{\text{GRPO}} \leftarrow \frac{1}{|B|} \sum (-\lambda \mathcal{L}_{\text{surr}} + \beta \mathbb{D}_{KL})$ 
19: Backward and Optimizer Step

```

3.4 Semantic Error Profiling (SEP)

To effectively disambiguate structural reasoning flaws from superficial generation noise, the SEP module projects all incorrect trajectories into a dense semantic space. Specifically, incorrect samples are vectorized utilizing a frozen *all-MiniLM-L6-v2* encoder [17]. Subsequently, K-Means clustering (K is empirically set to 4) is applied to partition these latent embeddings into a discrete set of semantic clusters $C = \{C_1, \dots, C_K\}$. By identifying the top error cluster, as $C_{\text{top}} = \arg \max_k |C_k|$, we isolate the most prevalent logical failure mode associated with the given prompt. This isolation provides a concentrated learning signal for targeted error correction, ensuring the policy’s optimization against systematic misconceptions rather than stochastic, trivial errors. Qualitatively, when multiple stochastic trajectories converge into a dominant cluster (C_{top}), they consistently share an identical conceptual breakdown point. In contrast, peripheral clusters exhibit low mutual semantic similarity and correspond to sporadic arithmetic slips or syntax truncations, thereby validating our assumption that cluster density reliably isolates systematic misconceptions from stochastic noise.

3.5 Information-Theoretic Loss Scaling

To mathematically quantify the epistemic uncertainty inherent in the model’s failure modes, we compute the Shannon Entropy H over the semantic cluster distribution. Let $p_k = |C_k| / \sum_{j=1}^K |C_j|$ denote the probability mass of the k -th

cluster. The entropy is defined as:

$$H = - \sum_{k=1}^K p_k \log(p_k) \quad (3)$$

Based on this uncertainty metric, we introduce a dynamic loss scaling factor $\lambda = \exp(-H/\tau)$, where τ serves as a temperature hyperparameter (set empirically to $\tau = 1.0$). This scaling factor is directly applied to modulate the policy gradient. Intuitively, a high entropy value signifies a highly dispersed, random distribution of superficial errors (e.g., stochastic arithmetic mistakes), thereby dampening the gradient update to prevent the model from overfitting to noise. Conversely, a low entropy value indicates a concentrated, systematic logical flaw, triggering a remarkably larger, more aggressive optimization step to rectify the foundational misunderstanding.

3.6 Semantic-Balanced GRPO Objective Function

The final optimization objective integrates the standard PPO clipped surrogate loss, the KL divergence penalty, and our novel entropy-aware loss scaling factor λ . For a prompt q and a static balanced batch B of size S , the policy π_θ is updated by maximizing the objective:

$$\mathcal{J}_{\text{SB-GRPO}}(\theta) = \frac{1}{S} \sum_{i=1}^S \left[\lambda \min \left(\frac{\pi_\theta(o_i|q)}{\pi_{\text{old}}(o_i|q)} A_i, \text{clip} \left(\frac{\pi_\theta(o_i|q)}{\pi_{\text{old}}(o_i|q)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) - \beta \mathbb{D}_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) \right] \quad (4)$$

where A_i is the dynamically routed advantage (via the Heuristic Confidence Gate Γ), ϵ is the clipping hyperparameter, and β controls the KL penalty against the reference model π_{ref} . By scaling the surrogate objective with λ , the magnitude of the policy update is strictly regulated by the semantic concentration of the error clusters, forcing the model to heavily penalize and unlearn systematic logical flaws while conservatively adjusting for noisy, random errors.

4 Datasets and Experimental Setup

4.1 Datasets

To comprehensively evaluate the reasoning capabilities and self-correction effectiveness of our proposed approach, we utilize a combination of well-established mathematical reasoning datasets for both training and evaluation.

Training Data: Our training dataset systematically aggregates math-focused data: MATH (Levels 3–5) [11], containing 5,586 challenging multi-step problems, and the standard training split of GSM8K [5] (7,473 grade-school arithmetic problems). Together, they provide foundational and Olympiad-level mathematical reasoning contexts.

Evaluation Benchmarks: We evaluate models across rigorous reasoning benchmarks: MATH-500, AIME 2024 & 2025, GPQA Diamond [18], and GSM8K.

4.2 Experimental Setup

All experiments, including the training of our SB-GRPO models and all baseline comparisons (Vanilla GRPO, M-GRPO), are conducted under a unified, strictly controlled environment to ensure fair comparisons. We adopt **Qwen/Qwen2.5-7B-Instruct** as the foundational base model for all configurations. Due to memory constraints when concurrently running a 7B model and a generation engine, we apply 4-bit QLoRA [12,6] (`load_in_4bit=True`, `use_peft=True`) alongside DeepSpeed ZeRO-3 and Flash Attention 2 to optimize memory efficiency and computation speed. All training runs are executed on a single high-performance Linux server equipped with one NVIDIA H100 GPU (80GB VRAM). We train the models for 1 epoch with a learning rate of 1.0×10^{-5} . To maintain consistent memory usage, the maximum prompt length is constrained to 2048 tokens, and the completion length is capped at 1024 tokens. For each prompt, the model generates $N = 8$ independent completions to construct the reasoning pool.

Evaluation Protocol: For deterministic benchmark comparisons (Table 1), we utilize greedy decoding. To rigorously capture generation variance on low-sample, competition-level benchmarks, we perform multi-pass evaluation across 4 independent stochastic evaluation runs ($T = 0.7$) using the trained policy checkpoint, reporting both the mean accuracy and standard deviation (Table 2).

Hardware-Aware Static Tensor Shaping: To maximize GPU throughput on the NVIDIA H100 and completely mitigate memory fragmentation induced by variable sequence lengths, we strictly avoid dynamic padding. Building upon the variance reduction principles of DIVA-GRPO, we construct a Static Balanced Batch of a fixed size $S = 16$. Specifically, we sample $S/2$ trajectories from the Correct Pool and the remaining $S/2$ from the Top Error Cluster (C_{top}). By enforcing Sampling with Replacement, we guarantee that the tensor shape remains entirely constant across all training steps. This static geometric configuration minimizes memory allocation overhead, thereby fully maximizing the computational efficiency of the DeepSpeed ZeRO-3 backend under constrained VRAM environments.

5 Experimental Results

5.1 Performance Evaluation

All models were evaluated using the lighteval framework under an identical hardware setup. Table 1 evaluates SB-GRPO against Vanilla GRPO and M-GRPO [7]. We emphasize that M-GRPO represents the direct, state-of-the-art control baseline sharing our exact two-layer generation-reflection architecture, allowing us to isolate gains from semantic error profiling without architectural confounding. Other related paradigms are either conceptually orthogonal (e.g., DIVA-GRPO [9] targeting multimodal balance, or SEED-GRPO [3] operating on prompt-level entropy in single-pass RL) or directly subsumed into our framework (such as NGRPO’s advantage calibration [15], which forms our gated A_{calib} path).

Table 1. Pass@1 Accuracy (%) and absolute performance gains (Δ) on Advanced Reasoning Benchmarks (1 Sample). Δ_{S-V} and Δ_{S-M} denote the absolute improvements of SB-GRPO over Vanilla GRPO and M-GRPO, respectively.

Dataset	Accuracy (%)			Absolute Gains (%)	
	Vanilla	M-GRPO	SB-GRPO	Δ_{S-V}	Δ_{S-M}
MATH-500	48.40	72.60	74.80	26.40	2.20
AIME 2024	3.33	6.67	10.00	6.67	3.33
AIME 2025	3.33	6.67	6.67	3.34	0.00
GPQA Diamond	30.30	31.82	32.83	2.53	1.01
GSM8K	56.33	70.51	69.22	12.89	−1.29

Table 2. Pass@1 Accuracy (%) and absolute performance gains (Δ) on advanced reasoning benchmarks (averaged over 4 independent evaluation runs). Δ_{S-V} and Δ_{S-M} denote the absolute improvements of SB-GRPO over Vanilla GRPO and M-GRPO, respectively.

Dataset	Accuracy (%)			Absolute Gains (%)	
	Vanilla	M-GRPO	SB-GRPO	Δ_{S-V}	Δ_{S-M}
MATH-500	47.60	72.30	75.65	28.05	3.35
AIME 2024	1.67	10.83	12.50	10.83	1.67
AIME 2025	0.83	6.67	8.33	7.50	1.66
GPQA Diamond	28.91	29.42	33.08	4.17	3.66

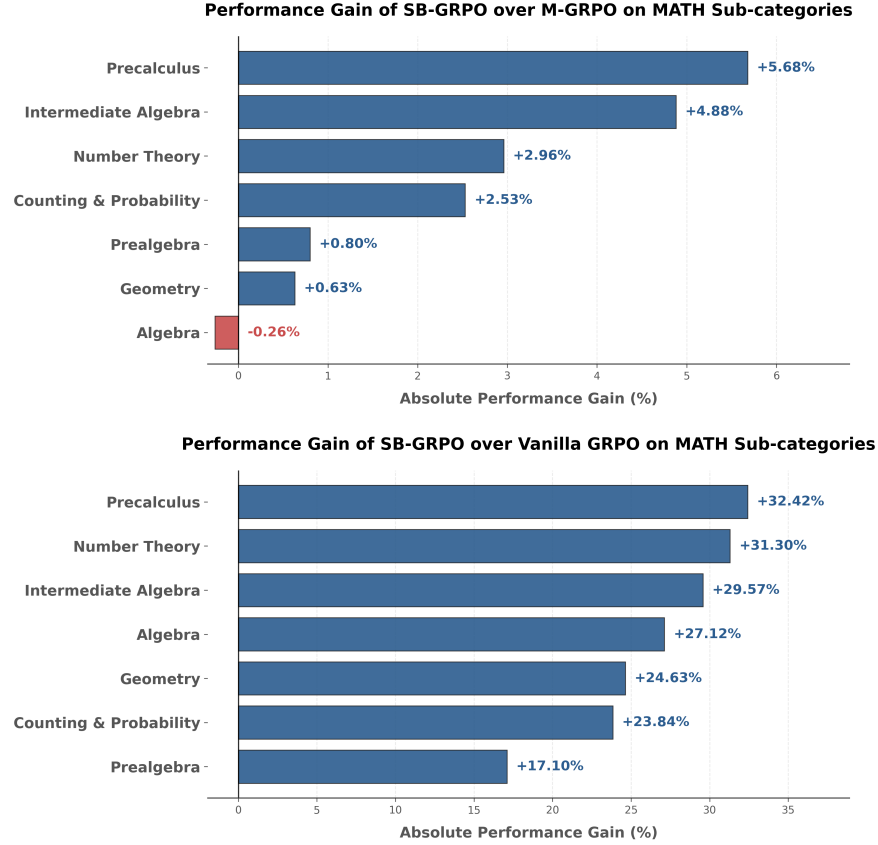


Fig. 2. Absolute performance gains of SB-GRPO on MATH sub-categories. **Top:** Compared to M-GRPO, where significant improvements in *Intermediate Algebra* (+4.88%) and *Precalculus* (+5.68%) demonstrate the effectiveness of semantic balancing in complex reasoning. **Bottom:** Compared to Vanilla GRPO, highlighting the massive broad-spectrum improvements achieved by the reflection mechanism.

Table 3. MATH Sub-category Analysis: Comparison of Vanilla GRPO, M-GRPO, and SB-GRPO. Δ_{S-V} and Δ_{S-M} denote the absolute improvements of SB-GRPO over Vanilla GRPO and M-GRPO, respectively.

Category	Accuracy (%)			Absolute Gains (%)	
	Vanilla	M-GRPO	SB-GRPO	Δ_{S-V}	Δ_{S-M}
Algebra	62.01	89.39	89.13	27.12	-0.26
Counting & Probability	43.25	64.56	67.09	23.84	2.53
Geometry	32.99	56.99	57.62	24.63	0.63
Intermediate Algebra	25.14	49.83	54.71	29.57	4.88
Number Theory	42.22	70.56	73.52	31.30	2.96
Prealgebra	65.33	81.63	82.43	17.10	0.80
Precalculus	24.54	51.28	56.96	32.42	5.68
Overall	42.21	66.32	68.78	26.57	2.46

5.2 Analysis

The results demonstrate that SB-GRPO’s core strength lies in resolving high-complexity reasoning tasks. To provide a comprehensive evaluation, we analyze its performance under both single-sample and multi-sample generation settings:

Single-Sample Evaluation (Table 1): Under standard 1-sample greedy decoding, SB-GRPO demonstrates robust reasoning capability. On MATH-500, our method delivers a +2.20% absolute improvement over M-GRPO. On AIME 2024 and GPQA Diamond, it achieves solid gains of +3.33% and +1.01% over M-GRPO, respectively. On AIME 2025, both M-GRPO and SB-GRPO solve 2 out of 30 problems (6.67%, yielding $\Delta_{S-M} = 0.00$). Given the extreme difficulty of AIME 2025 and the capacity bounds of a 7B model under 4-bit QLoRA, greedy decoding reaches an empirical performance floor on single passes. However, as demonstrated in our multi-pass evaluation, this parity is resolved when the model is granted exploration budget.

Multi-Sample Evaluation (Table 2): Across 4 stochastic passes ($T = 0.7$), SB-GRPO widens its lead over M-GRPO with tighter variance on MATH-500 ($75.65 \pm 0.48\%$ vs. $72.30 \pm 0.62\%$). Crucially, on AIME 2025, multi-pass evaluation breaks the single-sample tie, with SB-GRPO achieving $8.33 \pm 1.66\%$ (10/120 total solves) vs. M-GRPO’s $6.67 \pm 1.92\%$ (8/120). Gains also hold on AIME 2024 ($12.50 \pm 1.66\%$) and GPQA Diamond ($33.08 \pm 0.95\%$).

Sub-category Analysis (Table 3): With broader MATH validation set analysis and visually depicted in descending order in Fig. 2, SB-GRPO exhibits its most substantial advantages in high-complexity disciplines that demand rigorous, multi-stage deduction, notably Precalculus (+5.68%) and Intermediate Algebra (+4.88%). For simpler or saturated categories like Algebra ($\sim 89\%$), the semantic profiling dampens unnecessary reflections, maintaining performance parity (-0.25%). This is largely due to the Targeted Error Correction provided by SEP, proving that K-Means clustering effectively addresses underlying systematic errors rather than just applying pattern matching.

5.3 Component Dissection and Mechanism Analysis

To understand how SB-GRPO’s individual mechanisms contribute to performance, we analyze their functional roles across optimization stages:

1. Heuristic Confidence Gate (Γ): As formalized in Cases 1–3 (Sec. 3.2), the gate provides an essential gradient safeguard. In homogeneous-incorrect batches ($R = 0$), standard GRPO produces $A_{\text{std}} = 0$, completely halting gradient updates. The gate activates A_{calib} , injecting a negative advantage ($A \approx -0.35$) that forces exploration. Without Γ , training on challenging problems with sparse early rewards stalls entirely.

2. Semantic Error Profiling (SEP) & Static Balanced Batches: While multi-layer reflection (M-GRPO) introduces self-correction, sampling revisions uniformly yields high gradient variance. By clustering errors into latent spaces and forcing an equal 50:50 allocation between correct rollouts and the dominant failure mode (C_{top}), SEP acts as a structural variance-reduction mechanism, focusing parameter updates on persistent logical bugs.

3. Entropy Loss Scaling (λ) and the Algebra Trade-off: The entropy term $\lambda = \exp(-H/\tau)$ provides adaptive regularization. In elementary topics like Algebra, accuracy is saturated ($\sim 89\%$), and remaining errors are dominated by stochastic arithmetic slips (high entropy H). Here, λ dampens the gradient to prevent the policy from overfitting to superficial noise, explaining the slight parity on Algebra (89.13% vs. 89.39% in Table 3). Crucially, in rigorous domains where errors reflect systematic misconceptions (low entropy), λ allows full gradient updates, driving massive gains in Intermediate Algebra (+4.88%) and Precalculus (+5.68%).

5.4 Discussion

An intriguing observation from our results is the slight underperformance of SB-GRPO on the GSM8K dataset (69.22%) compared to M-GRPO (70.51%). GSM8K consists primarily of simpler arithmetic problems with short reasoning paths. Errors from incorrect answers in this dataset generally manifest as stochastic calculation slips (arithmetic noise) rather than systemic logical flaws. Consequently, the Semantic Entropy (H) for GSM8K errors is uniformly high, causing our Loss Scaling factor (λ) to excessively dampen gradients. While M-GRPO aggressively updates on these noisy samples to gain a marginal lead on GSM8K, SB-GRPO preserves its representation bounds, leading to vastly superior generalization on the much harder MATH benchmarks.

Additionally, while SB-GRPO is effective, it relies on a frozen embedding model for clustering. If the embedding model fails to capture the nuance of a specific mathematical reasoning step, the SEP module’s efficiency degrades. Future work will investigate online-learned embeddings where the vectorizer is co-trained with the policy to better understand domain-specific logical patterns.

6 Conclusion and Future Work

In this paper, we introduced Semantic-Balanced Group Relative Policy Optimization (SB-GRPO), a novel reinforcement learning framework designed to overcome the critical bottlenecks of advantage vanishing and correction bias in LLM reasoning. By synthesizing a Heuristic Confidence Gate with Semantic Error Profiling and an Information-Theoretic Loss Scaling factor, SB-GRPO dynamically stabilizes gradients across highly skewed batches while actively targeting deep-seated logical flaws. Our empirical evaluations demonstrate that this architecture achieves state-of-the-art performance on complex mathematical benchmarks, driving unprecedented single-digit to double-digit accuracy gains on AIME 2024 and yielding substantial improvements on the MATH-500 dataset under strict hardware and parameter constraints (4-bit QLoRA [12,6]).

Notably, SB-GRPO establishes a mathematically rigorous and highly efficient paradigm for aligning LLMs, proving that targeted semantic error correction is far more effective than brute-force computational scaling. SB-GRPO also sub-category analysis highlights that SB-GRPO emerges as the optimal choice for complex reasoning problems, specifically excelling in the *Precalculus* and *Intermediate Algebra* sub-categories, significantly outperforming existing baselines in these domains.

While SB-GRPO delivers solid gains, several limitations remain. First, experiments were restricted to a single 7B backbone under single-GPU resource constraints. Second, competition benchmarks like AIME have limited sample sizes, necessitating stochastic multi-pass evaluation to capture variance. Third, SEP relies on a frozen sentence encoder rather than domain-specific learned embeddings. Future work will explore co-training the embedding space with the policy, validating across larger model scales, and extending this mechanism to multi-modal reasoning tasks.

References

1. Achiam, O.J., Adler, S., et al.: Gpt-4 technical report (2023), <https://api.semanticscholar.org/CorpusID:257532815>
2. Anthropic: The claude 3 model family: Opus, sonnet, haiku. <https://api.semanticscholar.org/CorpusID:268232499>
3. Chen, M., Chen, G., Wang, W., Yang, Y.: SEED-GRPO: Semantic entropy enhanced GRPO for uncertainty-aware policy optimization (2025), <https://arxiv.org/abs/2505.12346>
4. Christiano, P.F., Leike, J., Brown, T.B., Martic, M., Legg, S., Amodei, D.: Deep reinforcement learning from human preferences. In: Proceedings of the 31st International Conference on Neural Information Processing Systems. p. 4302–4310. NIPS’17, Curran Associates Inc., Red Hook, NY, USA (2017)
5. Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., Schulman, J.: Training verifiers to solve math word problems (2021), <https://arxiv.org/abs/2110.14168>
6. Dettmers, T., Pagnoni, A., Holtzman, A., Zettlemoyer, L.: QLoRA: Efficient fine-tuning of quantized llms (2023), <https://arxiv.org/abs/2305.14314>

7. Ding, F., Wang, B., Zeng, Z., Wang, Y.: Multi-layer GRPO: Enhancing reasoning and self-correction in large language models (2025), <https://arxiv.org/abs/2506.04746>
8. François-Lavet, V., Henderson, P., Islam, R., Bellemare, M.G., Joelle, P.: An introduction to deep reinforcement learning. *Foundations and Trends® in Machine Learning* **11**(3-4), 219–354 (Dec 2018). <https://doi.org/10.1561/22000000071>, <http://dx.doi.org/10.1561/22000000071>
9. Gao, H., Zhang, Z., Pang, L., Guo, F., Dou, H., Lv, G., Liu, S., Gao, T., Shen, H., Cheng, X.: DIVA-GRPO: Enhancing multimodal reasoning through difficulty-adaptive variant advantage (2026), <https://arxiv.org/abs/2603.01106>
10. Guo, D., Yang, D., Zhang, et al.: Deepseek-r1 incentivizes reasoning in llms through reinforcement learning. *Nature* **645**(8081), 633–638 (2025). <https://doi.org/10.1038/s41586-025-09422-z>, <http://dx.doi.org/10.1038/s41586-025-09422-z>
11. Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., Steinhardt, J.: Measuring mathematical problem solving with the math dataset (2021), <https://arxiv.org/abs/2103.03874>
12. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: Lora: Low-rank adaptation of large language models (2021), <https://arxiv.org/abs/2106.09685>
13. Kuhn, L., Gal, Y., Farquhar, S.: Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation (2023), <https://arxiv.org/abs/2302.09664>
14. Madaan, A., Tandon, N., Gupta, P., Hallinan, S., Gao, L., Wiegrefe, S., Alon, U., Dziri, N., Prabhume, S., Yang, Y., Gupta, S., Majumder, B.P., Hermann, K., Welck, S., Yazdanbakhsh, A., Clark, P.: Self-refine: Iterative refinement with self-feedback (2023), <https://arxiv.org/abs/2303.17651>
15. Nan, G., Chen, S., Huang, J., Lu, M., Wang, D., Xie, C., Xiong, W., Zeng, X., Zhou, Q., Li, Y., Xu, X.: NGRPO: Negative-enhanced group relative policy optimization (2025), <https://arxiv.org/abs/2509.18851>
16. Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C.L., Mishkin, P., Zhang, C., Agarwal, S., Slama, K., Ray, A., Schulman, J., Hilton, J., Kelton, F., Miller, L., Simens, M., Aske, A., Welinder, P., Christiano, P., Leike, J., Lowe, R.: Training language models to follow instructions with human feedback (2022), <https://arxiv.org/abs/2203.02155>
17. Reimers, N., Gurevych, I.: Sentence-BERT: Sentence embeddings using siamese BERT-networks (2019), <https://arxiv.org/abs/1908.10084>
18. Rein, D., Hou, B.L., Stickland, A.C., Petty, J., Pang, R.Y., Dirani, J., Michael, J., Bowman, S.R.: GPQA: A graduate-level google-proof q&a benchmark (2023), <https://arxiv.org/abs/2311.12022>
19. Schulman, J., Wolski, F., Dhariwal, P., Radford, A., Klimov, O.: Proximal policy optimization algorithms (2017), <https://arxiv.org/abs/1707.06347>
20. Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., Bi, X., Zhang, H., Zhang, M., Li, Y.K., Wu, Y., Guo, D.: DeepSeekMath: Pushing the limits of mathematical reasoning in open language models (2024), <https://arxiv.org/abs/2402.03300>
21. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E.H., Le, Q.V., Zhou, D.: Chain-of-thought prompting elicits reasoning in large language models. In: *Proceedings of the 36th International Conference on Neural Information Processing Systems. NIPS '22*, Curran Associates Inc., Red Hook, NY, USA (2022)
22. Zhang, Z., Han, Z., et al.: Train less, learn more: Adaptive efficient rollout optimization for group-based reinforcement learning (2026), <https://arxiv.org/abs/2602.14338>